

100% Evidence-Based Cross-System Medical Platform

SYSTEM ARCHITECTURE & VERIFICATION BLUEPRINT

Building a platform that compares modern medicine (Allopathy) with traditional medical frameworks (Ayurveda, Siddha, Naturopathy) while strictly adhering to a 100% evidence-based paradigm requires a rigorous methodology. The primary challenge stems from structural asymmetries in data: modern pharmaceutical drugs follow established, standardized regulatory pathways (RCTs, meta-analyses), whereas traditional treatments frequently rely on historic scripts, observational reports, or smaller trials. To preserve total scientific integrity, the platform must clearly distinguish and categorize these layers using an objective evidence engine.

1. The Hierarchical Evidence Engine

To eliminate systemic bias while upholding scientific objectivity, the platform must grade every clinical, botanical, or therapeutic claim using a rigorous taxonomy adapted from the Oxford Centre for Evidence-Based Medicine (CEBM).

Grade	Confidence Level	Required Empirical Criteria	Application Example
GRADE A	High / Definitive	Double-blind, multi-center Randomized Controlled Trials (RCTs), Systematic Reviews, and Cochrane-style Meta-analyses.	Metformin for Type 2 Diabetes; DASH Diet/ Potassium supplementation for Hypertension.
GRADE B	Moderate / Emerging	Small-scale clinical trials, cohort studies, well-designed case-control studies, and rigorous open-label trials.	Standardized <i>Curcumin</i> extracts for joint inflammation; <i>Ashwagandha</i> for stress-cortisol regulation.
GRADE C	Traditional / Mechanistic	Pre-clinical laboratory data (in-vitro/ in-vivo animal models), official government pharmacopeias, and historic textual consensus.	Specific multi-herbal Siddha formulations or specialized Naturopathic hydrotherapy protocols lacking human trials.

2. Data Harmonization & Schema Architecture

Traditional systems map disease vectors through systemic frameworks (e.g., metabolic imbalances or *Doshas*) rather than specific modern biochemical markers. The relational schema must link these disparate clinical terminologies via a centralized, cross-referenced table, mapping each claimed curative intervention directly to its empirical verification profile.

```
-- Core Diseases & Taxonomy Directory
CREATE TABLE medical_conditions (
    condition_id SERIAL PRIMARY KEY,
    icd_11_code VARCHAR(15) UNIQUE, -- International Classification of Diseases mapping
    standard_name VARCHAR(15) NOT NULL, -- e.g., "Type 2 Diabetes"
    ayurvedic_equivalent VARCHAR(255), -- e.g., "Prameha / Madhumeha"
    siddha_equivalent VARCHAR(255), -- e.g., "Madhumegam"
    pathophysiology_summary TEXT
);

-- Interventions & Empirical Verification Engine
CREATE TABLE treatment_evidence (
    evidence_id SERIAL PRIMARY KEY,
    condition_id INT REFERENCES medical_conditions(condition_id),
    medical_system VARCHAR(50) NOT NULL, -- 'Allopathy', 'Ayurveda', 'Siddha', 'Naturopathy'
    intervention_name VARCHAR(255) NOT NULL, -- e.g., 'Lisinopril' vs 'Sarpagandha'
    is_pharmacological BOOLEAN DEFAULT TRUE, -- TRUE = Medicated, FALSE = Non-Medicated
    (Lifestyle/Diet)
    evidence_grade CHAR(1) CHECK (evidence_grade IN ('A', 'B', 'C')),
    study_methodology VARCHAR(100), -- e.g., 'Double-blind RCT', 'Systematic Review'
    sample_size INT,
    registry_identifier VARCHAR(100), -- PubMed PMID, DOI, CTRI ID, or AYUSH Ref
    source_url TEXT NOT NULL,
    clinical_outcome_summary TEXT NOT NULL,
    last_verified_date DATE DEFAULT CURRENT_DATE
);
```

3. Automated Evidence Ingestion Pipeline

To scale data curation, the platform utilizes specialized public medical and clinical APIs. The data collection layer follows an exact ingestion workflow:

Automated API Integration Matrix

- **Modern Allopathic Data:** Queried through the NCBI Entrez Utilities (PubMed/MEDLINE API) and ClinicalTrials.gov to fetch randomized control trials and meta-analyses.
- **Traditional Indian Medicine (AYUSH):** Extracted via the official AYUSH Research Portal and the DHARA (Digital Helpline for Ayurveda Research Articles) index to map validated traditional clinical assessments.
- **Integrative Clinical Registries:** Linked to the Clinical Trials Registry - India (CTRI) and the WHO International Clinical Trials Registry Platform (ICTRP) to capture emerging hybrid and comparative studies.

4. Editorial Protocol & Integrity Safeguards

Maintaining absolute evidentiary purity requires strict platform constraints:

1. **No Causal Equivalency:** If an Ayurvedic herb demonstrates a Grade B reduction in blood glucose, the interface must explicitly show it beside a Grade A Allopathic drug without implying they are interchangeable.
2. **Mandatory Risk Metrics:** Contraindications and multi-system drug interactions (e.g., an herbal remedy altering the liver clearance of an allopathic blood thinner) must be surfaced with equal priority to clinical efficacy.
3. **Dynamically Indexed Sources:** No claim can be published unless it directly references a live, verifiable digital identifier (PMID/DOI/Registry ID). Anecdotal or crowdsourced testimonials are systematically blocked by the schema.